

How does Convolutional Neural Network (CNN) compare to a RS statistical test in accurately and reliably detecting LSB steganography in PNG images?

Gullbrandsson, Martin

Pakos, Jan

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Supervisor: Johannes Olegård
Department of Computer
and Systems Sciences



**Stockholm
University**

Abstract

Steganalysis has typically relied on hand-crafted statistical means of discovering concealed data; however, in recent years the effort has largely shifted towards automated feature extraction techniques via deep learning. Reliable detection of such hidden content is of direct relevance to law enforcement and cybersecurity practitioners tasked with uncovering covert communication channels in digital forensic investigations. This thesis examines whether the newer Convolutional Neural Networks (CNNs) are more effective than traditional RS-analysis methods to detect Least Significant Bit (LSB) steganography utilized to hide data in lossless PNG image files. To evaluate these two detection methods, a quantitative experiment was performed using the BOSSBase 1.01 dataset on 4 different embedding techniques: Sequential LSB, Randomized LSB, Edge-Adapted LSB, and LSB Matching, at three different payload capacities of 25%, 50%, and 75%. The results of this investigation show that traditional RS-analysis or modern CNN methods can be optimal based on what embedding strategy has been used. RS-analysis was superior to CNN (SRNet) for both Sequential and Randomized LSB modifications across all examined payloads, achieving greater than 95% accuracy for both modified versions even at the lowest embedding rate of 25%. Conversely, for low payload Edge-Adaptive LSB, CNN demonstrated superior robustness with an accuracy of 79.12% versus the RS-analysis accuracy of 72.50% at a 25% payload. Finally, only the CNN was capable of detecting LSB Matching (with an accuracy of 91.36%), where the RS-analysis performed below chance level in detecting this embedding method. Therefore, these results demonstrate that traditional statistical techniques continue to be effective and sufficient for the basic Least Significant Bit substitution; however, deep learning techniques are needed for detecting Edge-Adaptive LSB and/or LSB Matching steganography.

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
AUC	Area Under the Curve
CNN	Convolutional Neural Network
DSR	Design Science Research
FPR	False Positive Rate
GNCNN	Gaussian-Neuron Convolutional Neural Network
LSB	Least Significant Bit
ML	Machine Learning
NLP	Natural Language Processing
PRNG	Pseudo-Random Number Generator
ROC	Receiver Operating Characteristic
RQ	Research Question
RS	Regular-Singular
SRNet	Steganalysis Residual Network
SQ	Sub-question
TPR	True Positive Rate
Bpp	Bits Per Pixel

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1

Introduction

In the age of widespread digital communication and increasing concerns about data privacy, the methods of concealing information have become vital in computer and systems sciences research. Steganography, derived from the Greek words *steganos* (covered) and *graphein* (writing), is defined as the practice of concealing information inside often-innocent digital media, such as images, for secret communication [10]. While cryptography encrypts information in order to make it unreadable, steganography is concerned with concealing the very existence of the information [10], making it valuable for scenarios where the attention drawn to encrypted messages would be undesirable.

Steganographic methods have a number of serious implications concerning security and forensics. Although they can be used for legal purposes, such as digital watermarking used for copyright protection, secure communication in countries that ban or restrict encryption, and privacy preservation, these methods can be used for other malicious tasks as well [13], which makes reliable detection a matter of broader societal interest. Research into steganalysis therefore contributes directly to the work of forensic specialists and cybersecurity practitioners tasked with uncovering hidden communication channels that would otherwise evade conventional inspection.

The countermeasure to steganography is steganalysis. An ongoing “arms race” has driven the development of numerous techniques aimed at reliably detecting the presence of hidden data in digital media. These detection methods fall primarily into two categories.

The first category consists of traditional statistical methods, such as Regular-Singular (RS) analysis, which detects disruptions in the spatial correlations and local smoothness that random LSB embedding introduces to pixel data [11, 23].

The second, more recent category employs machine learning—specifically deep learning. Models such as Convolutional Neural Networks (CNNs) are trained to detect the fine, high-dimensional patterns and noise artifacts introduced into stego images [26].

The advancements in deep learning research have resulted in a paradigm shift for steganalysis research with a tendency to regard Neural Networks as the norm for solving steganalytic detection tasks. However, this shift might neglect computational complexity and the baseline detection accuracies of older methods. Although the use of Convolutional Neural Networks (CNNs) with state-of-the-art performances for complex and adaptive steganographic attacks is compelling, their superiority over traditional methods for basic LSB embedding schemes within lossless images is still worthy of research.

This study compares the two approaches, and aims to provide a direct and empirical comparison of their respective effectiveness.

1.1 Delimitations

1.1.1 Image file format (PNG Only)

The study focuses on Least Significant Bit (LSB) steganography, which requires exact pixel value preservation. Lossy image formats, such as JPEG, destroy LSB information in the compression process and introduce artifacts that interfere with the specific statistical analysis methods being tested. Performing both steganography and steganalysis on such image formats requires significantly higher sophistication, and these are outside our scope.

1.1.2 Spatial Domain LSB Substitution

This research aims to compare the baseline effectiveness of a statistical method (RS-analysis) against a Deep Learning model (CNN) in the spatial domain. While complex, highly adaptive steganographic algorithms (e.g., HUGO [25], S-UNIWARD [18]) fall outside the scope of this comparison due to requiring entirely different steganalytic approaches, foundational spatial techniques, such as basic sobel based edge adapted steganography, is included to provide a comprehensive baseline.

1.1.3 Dataset and Color Channels

The BOSSBase dataset is widely used in the field of steganalysis research, ensuring reproducibility [2, 27]. The datasets images are in a lossless format (.png), and as the images in the dataset are in grayscale, the embedding artifacts are isolated within the luminance values.

1.1.4 Payload Granularity

This study opts for discrete payload sizes rather than a continuous spectrum. In order to answer the sub-question regarding changes as payload decreases, the study tests fixed embedding rates (25%, 50%, and 75% of image embedding capacity for respective steganographic method) to provide clear comparative bins for the True Positive Rate and overall detection accuracy analysis.

1.2 Research Question

This study is an empirical comparison, intent on investigating the effectiveness of traditional statistical steganalysis versus modern deep learning classification in detecting steganographic embedding.

This objective is condensed into the following **Research Question (RQ)**:

RQ: How does Convolutional Neural Network (CNN) compare to a RS-Analysis statistical test in accurately and reliably detecting LSB steganography in PNG images?

With the help of the following **Sub-Question (SQ)**:

SQ : How does the detection accuracy of each method change as the size of the hidden payload decreases?

2

Extended Background

2.1 Literature Review

2.1.1 Fundamentals of LSB Steganography

LSB embedding stands out as one of the most popular and accessible steganographic techniques. LSB steganography replaces the value of a least significant bit of a pixel with the bit from the hidden message, as described by Chan and Cheng [4]. For instance, in an 8-bit grayscale image, modifying the least significant bit of a pixel value from 10110110 (182) to 10110111 (183) results in a change of only one intensity level (see Figure 2.1), which is essentially undetectable by the human eye. The advantage of LSB insertion lies in its simplicity and high embedding capacity; however, this embedding process also introduces statistical abnormalities that can be utilized in the detection process.

The image format used to contain the hidden information plays a vital role in both the effectiveness of the steganographic embedding and its detectability. While not the only lossless format, PNG (Portable Network Graphics) images are highly relevant to LSB steganography given their characteristics as a lossless compressible file format. While JPEG is a useful format, it is lossy and often destroys hidden data when compression artifacts are introduced [8]. PNG images and their deflate compression algorithm preserve pixel values exactly, thus maintaining an ideal medium for LSB

steganography because the image maintains the exact bit values required for message recovery.

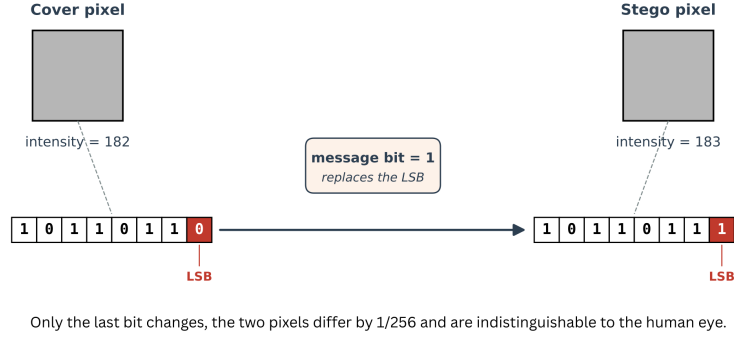


Figure 2.1: LSB Illustration.

2.1.2 Traditional Steganalysis

Fridrich, Goljan, and Du [11] introduced the RS (Regular-Singular) steganalysis method to detect LSB steganography in grayscale images. Unlike histogram attacks (like Chi-Square) that look for global frequency anomalies, RS analysis uses spatial correlations between groups of pixels to estimate the initial cover image.

RS analysis works by dividing the image into disjoint groups of n adjacent pixels (e.g., 2×2 blocks). A discrimination function, f , is defined to measure the “noisiness” or variation within each group. As defined by Fridrich, Goljan, and Du [11], for a group of pixels $G = (x_1, x_2, \dots, x_n)$, the discrimination function is calculated as the sum of absolute differences between adjacent pixels:

$$f(x_1, x_2, \dots, x_n) = \sum_{i=1}^{n-1} |x_{i+1} - x_i| \quad (2.1)$$

By applying a flipping operation (mask M) to the LSBs of the pixels in a group, the block’s noisiness changes. Based on how the discrimination function value changes after flipping (denoted as $F_M(G)$), groups are categorized into:

- **Regular (R):** Flipping increases the noisiness ($f(F_M(G)) > f(G)$).
- **Singular (S):** Flipping decreases the noisiness ($f(F_M(G)) < f(G)$).

- **Unusable (U):** Flipping does not change the noisiness ($f(F_M(G)) = f(G)$).

Statistical symmetry is the core premise of RS analysis. In an image with no embedding, the number of Regular Groups (R_M) should be approximately equal to the number of Regular groups obtained using an inverse mask (R_{-M}); the case is similar for Singular groups:

$$R_M \approx R_{-M} \quad \text{and} \quad S_M \approx S_{-M} \quad (2.2)$$

The natural balance is disturbed by LSB embedding, and the disturbance grows as the payload increases. As the size of this modified payload increases, the difference between R_M and S_M will become smaller. This process continues until the image becomes completely randomized; at that point, R_M and S_M converge. Since RS Analysis measures absolute deviation from this perfect symmetry, it not only allows for detection of steganography but also allows for a more accurate estimation of embedded message length than earlier first-order statistical tests.

2.1.3 Deep Learning in Steganalysis

The limitations of handcrafted statistical features, such as the rigid assumptions regarding spatial correlation in RS-analysis, paved the way for deep learning approaches. While RS-analysis relies on explicit knowledge of the embedding process, deep learning (DL) methods can automatically learn complex patterns that distinguish cover images from stego-images. This capability gives DL methods an advantage, as a Convolutional Neural Network (CNN) can learn to identify steganographic artifacts from raw data that remain statistically invisible to fixed-model testing methods [9].

The field has experienced a paradigm shift from spatial-domain analysis toward adaptive domain techniques driven by artificial intelligence [1]. The primary method that has emerged is the use of CNNs. However, the implementation of CNNs for steganalysis provides unique challenges compared to traditional computer vision tasks. In a classic example of image recognition (e.g., recognizing a cat in a photo), the object being identified is the dominant signal.

Steganographic modifications, conversely, are designed to be imperceptible, hidden behind the “strong” noise of the image content [26]. Standard

CNN techniques usually fail to detect these weak signals because the high-amplitude content suppresses them. To address this, Qian et al. [26] proposed a customized deep learning model known as the Gaussian-Neuron CNN (GNCNN).

The GNCNN introduces a pre-processing layer designed to suppress image content and enhance the signal-to-noise ratio of the steganographic embedding. A high-pass filter is applied before the data enters the convolutional layers, effectively removing the “cover image” and leaving the “noise residual” where steganographic artifacts reside. This is similar to the residual calculation step in traditional steganalysis but is integrated directly into the learning pipeline.

The key benefit of using CNNs over traditional statistical methods, such as RS-analysis, is that the CNN architecture combines both feature extraction and classification into one unified structure. In traditional methods, feature extraction is completed first, and those features are used to train the classifier independently; with CNNs, both feature extraction and classification occur in concert using backpropagation [26].

2.1.4 Sensitivity and Robustness

Researching varying payload size highlights the challenge of detection sensitivity. RS-Analysis relies on comparing variations within block statistics (based on R_M versus S_M). As the payload decreases, the difference becomes negligible. At low payload size (below 0.05 bpp), the modification to pixel groups does not create enough signal to differentiate the image from natural noise. While statistical approaches struggle to create an accurate threshold in these low-payload situations, deep learning models have shown superior performance [1].

Combating the false positive rate is also vital for operational security. A key weakness of statistical tests is the occurrence of false alarms when viewing noisy images or those with distinct textures. Scanned photographs or images altered using specific filters often possess statistical properties that confuse RS steganalysis, incorrectly classifying clean images as stego-images [10]. CNNs typically offer better generalizability. After being trained on large datasets, deep learning models can learn to detect the specific artificial noise introduced by LSB embedding, reducing the false positive rate compared to rigid statistical tests [26, 1].

3

Method

To address the research question, this study employs a quantitative experimental setup. In line with the description and definition of an experiment by Denscombe [6], this study manipulated independent variables in order to measure their impact on the dependent variables. This was done for each steganalysis method (RS-Analysis and CNN), and the yielded observations were compared in order to draw conclusions concerning the relative effectiveness of the steganalysis methods.

In this study, the steganographic embedding method and the payload size were used as independent variables, and detection accuracy, true positive rate, and false positive rate as the dependent variables.

3.1 Ethical Considerations

This study relies entirely on the processing of pre-existing digital data and does not involve human participants. Thus, the ethical concerns regarding informed consent, anonymity, and voluntary participation are not applicable [6]. The BOSSBase 1.01 dataset [2] is a standard academic image library recognized in forensics research. No personal or sensitive data was collected or generated during the experimental process.

The key ethical consideration of this study relates to the “dual-use” of steganography and steganalysis. While steganography can be applied for ethical purposes like protecting privacy and copyrights; it can also be used for

malicious purposes like data exfiltration attacks and command and control communication in cyber-attacks. Studying steganalysis, or techniques for detecting steganographic communications, is a positive contribution as it helps forensic specialists in detecting hidden communication channels. Instead of using the research for developing new embedding algorithms that are difficult to be detected, this study strictly focuses on the detection of LSB steganography, which is an example of good security research ethics.

To ensure scientific integrity, this study takes the importance of replicability into consideration. Using a public data set with standard evaluation criteria (True Positive Rate, True Negative Rate, False Positive Rate, False Negative Rate, Accuracy) helps ensure that the comparison between the CNN and RS-Analysis is sound, so that the capability of the deep learning approach is not misrepresented for security applications.

Further ethical implications of this study are discussed in section 5.

3.2 Experimental Design

The conduct of the experiment can be examined as a four-step process.

1. Cover Image Selection and Preparation.
2. Stego-Image Embedding.
3. Implementation of Steganalysis
4. Performance Evaluation.

3.2.1 Cover Image Preparation

Dataset

BOSSBase 1.01 was selected due to its established prevalence in the field of steganography research [27]. While the original dataset images were provided in Portable Graymap (.PGM) format, these images were subsequently converted into Portable Network Graphics (.PNG) format. This conversion maintained the lossless nature of the images while providing a closer resemblance to real-world digital environments, where PNG files are significantly more common [30].

3.2.2 Stego-Image Generation

Payload Sizes (Embedding Ratios)

For the size of the embedding message 25% ($r = 0.25$), 50% ($r = 0.50$), and 75% ($r = 0.75$) of the embedding capacity were chosen.

LSB Methods

Four different LSB-based steganographic methods of varying sophistication were selected for comparative testing. This was done with the goal of increasing the generalizability of the assessment. These were implemented in Python scripts from the theoretical sources listed in Table 3.1.

METHOD	DESCRIPTION	SOURCES
Sequential LSB	LSB embedding in its simplest form.	[4]
Randomized LSB	Use of PRNG for selecting embedding locations.	[28]
LSB Matching (± 1)	Randomly adds or subtracts 1 from the pixel value instead of replacing it.	[19]
Sobel Edge Adaptive LSB	Identifies “noisy” pixels for embedding.	[22, 29]

Table 3.1: LSB Methods in this Study

3.2.3 Steganalysis Implementation

RS-Analysis

This steganalysis method was implemented in Python, adhering to the specifications outlined by Fridrich, Goljan, and Du [11].

Convolutional Neural Network (CNN)

For the ML model, SRNet (Steganalysis Residual Network) [3] was chosen. This was motivated by its architecture, which aims to extract noise residuals, where the hidden messages might reside, in early layers, before they become suppressed in successive layers [31], and its documented performance detecting stego images embedded with advanced steganographic methods [3].

SRNet Implementation and Training Details

To ensure full reproducibility, the specific hyperparameters and training configurations utilized for the SRNet model are detailed below.

Weight Initialization The convolutional layers of the network were initialized using a Xavier uniform distribution, with biases set to a constant value of 0.2. The fully connected (Linear) layers were initialized using a Normal distribution ($\mu = 0.0, \sigma = 0.01$) with biases set to 0.0.

Optimizer and Loss Function The network was trained using the Adamax optimizer [20], since Boroumand, Chen, and Fridrich [3] found it reliable with a fast convergence relative to other alternatives. The optimizer hyperparameters were configured to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with no weight decay applied. The final layer utilizes a LogSoftmax activation, and the model was optimized against a Negative Log-Likelihood Loss (NLLoss) function.

Training Pipeline and Augmentation A batch size of 20 was utilized during training. The initial learning rate was set to 0.001 and was decayed by a factor of 0.1 every 30 epochs. To increase robustness, data augmentation was applied to the training set by randomly rotating the input images by 90° .

Table 3.2: SRNet training configuration.

Parameter	Value
Input size	256×256 grayscale
Train / Val / Test	7000 / 1500 / 1500
Batch size	20
Optimizer	Adamax
Initial learning rate	1×10^{-3}
LR schedule	$\times 0.1$ every 30 epochs
Loss	NLLoss (LogSoftmax output)
Augmentation	Random rotation up to 90°
Framework	PyTorch 2.x

3.2.4 Performance Metrics

The results of the experiment were compiled into two tables with several relevant metrics, grouped by embedding technique.

3.3 Data and Code Availability

To ensure full transparency and to facilitate the reproducibility of the experimental pipeline, all custom scripts, model architectures, and data processing pipelines developed for this thesis have been made publicly available.

The primary repository containing the overall experimental pipeline and the steganographic embedding methods can be found in the main project repository [16]. The implementations of the steganalysis methods, SRNet [15] and RS-analysis [14], utilized pre-existing implementations as a base, which were forked and modified to fit this experiment. These are available as subrepositories within the main project repository and can be accessed independently. The creators of the original repositories are properly attributed within the respective forks.

Readers and researchers are encouraged to consult these repositories for exact code-level details regarding the hyperparameters, preprocessing steps, and evaluation loops utilized in this study.

3.4 Alternative Research Strategies and Methods

3.4.1 Choice of Dataset

Sedighi, Fridrich, and Cogan [27] concludes that the choice of cover images has a huge impact on the effectiveness of different steganographic and steganalysis methods. Evaluating effectiveness of these methods on the same cover image source might thus not yield a complete generalizable result. An alternative approach would be to have the cover image source as an independent variable, although this would enlarge the study considerably.

Another approach would be to focus the study on certain fixed scenarios where cover source and steganographic method are predefined to be the presumed optimal for each other, and then compare different steganalysis methods.

3.4.2 Other Steganographic Methods

This study does not include more modern steganographic methods like WOW, HUGO, MiPOD, or S-UNIWARD. This was done to limit the study,

but decreases its usability in a modern context. Another approach could include or focus solely on one or more of these methods.

3.4.3 Design Science Research

One possible alternative approach that might have been adopted is Design Science Research (DSR), which has been formally introduced by Chatterjee and Hevner [5]. In DSR, the focus is on developing and assessing a new artifact, such as a new detection algorithm, an improved CNN framework, or an integrated detection scheme based on statistics and deep learning, with the designed artifact being the key research contribution. In this process, knowledge creation occurs by repeated building and testing, followed by communicating design knowledge as the research outcome. DSR was taken into consideration but eventually rejected as the main method for use in the current thesis. This is due to the nature of the research question asked, which is comparative in its approach rather than constructive.

3.4.4 The Use of AI tools

To meet the requirements of academic integrity, the authors wish to acknowledge that AI tools were used in the process of writing this thesis in a supporting role. In particular, AI programs were used for: verifying the spelling and grammar in sentences, proposing rewrites aimed at enhancing the readability of sentences written by the authors, helping find and pinpoint source literature, which was then reviewed, validated, and referenced by the authors directly, and giving support in coding the experiment pipeline, involving debugging and syntax.

4

Results

RS estimates were computed for both clean and stego images across three embedding rates ($r \in \{0.25, 0.50, 0.75\}$) in order to evaluate the detectability of each embedding technique. For clean images, the mean RS estimated embedding rate was 0.0211 (actual rate was zero) with a standard deviation of 0.0842.

A binary classifier was constructed by thresholding the RS estimate at a value τ . An image whose estimated embedding rate met or exceeded τ was classified as stego; otherwise, it was classified as cover. This yields four outcomes: a stego image correctly classified as stego is a True Positive (TP), while one falling below τ is a False Negative (FN). A cover image correctly classified as cover is a True Negative (TN), while one meeting or exceeding τ is a False Positive (FP). The threshold τ was set to match the respective embedding rate tested (0.25, 0.50, or 0.75).

4.0.1 ROC & AUC

For both steganalysis algorithms described above, we obtain a score per image reflecting its likelihood of carrying concealed data. To turn this score into a binary decision, we must choose a threshold. An image with a score above the threshold is considered stego; otherwise, it is classified as clean. Setting the threshold is a balancing act: a lower threshold increases true positives (TP) at the expense of false positives (FP), while a higher threshold increases true negatives (TN) at the expense of false negatives (FN).

Table 4.1: Classification performance of RS analysis for different LSB embedding techniques and rates.

Strategy	Rate	AUC	Accuracy	Threshold τ	TP	TN	FP	FN
Sequential	0.25	0.982	95.4%	0.25	1441	1420	80	59
	0.50	0.998	98.6%	0.50	1482	1477	23	18
	0.75	0.999	99.8%	0.75	1497	1496	4	3
Random	0.25	0.986	97.0%	0.25	1469	1440	60	31
	0.50	0.999	99.5%	0.50	1497	1487	13	3
	0.75	0.999	99.9%	0.75	1499	1498	2	1
Adaptive	0.25	0.761	72.5%	0.25	1095	1081	419	405
	0.50	0.954	90.2%	0.50	1385	1321	179	115
	0.75	0.997	98.1%	0.75	1487	1456	44	13
Matching	0.25	0.516	53.3%	0.25	493	1106	394	1007
	0.50	0.504	51.9%	0.50	448	1108	392	1052
	0.75	0.514	53.6%	0.75	498	1110	390	1002

The ROC plot maps out this balance. For each threshold, it displays the ratio of TP to the sum of TP and FN (True Positive Rate, TPR) along the vertical axis, and FP to TN+FP (False Positive Rate, FPR) on the horizontal axis. A classifier with a perfect hit rate will pass exactly through the upper-left corner of the plot at (0,1). A classifier making random decisions will yield a 45-degree line. The Area Under Curve (AUC) summarizes the information in an ROC plot in a single numerical measure between zero and one. AUC corresponds to the probability that the classifier will assign a higher score to a randomly chosen stego image than to a randomly chosen clean image. Thus, an AUC of 1.0 implies that the two classes have no overlap, while 0.5 would mean that the classifier makes random decisions. Since AUC does not rely on a specific threshold, it permits direct comparison between classifiers with scores of different units, such as the RS estimates and CNN probabilities used in our experiment.

For the Sequential and Random strategies, detection performance is high across all three embedding rates, with AUC values exceeding 0.98 in every case. At an embedding rate of $r = 0.75$, both strategies yield near-perfect accuracy (99.8% and 99.9%, respectively). Even at the lowest rate of $r = 0.25$, RS analysis achieves AUC values of 0.982 (Sequential) and 0.986 (Random), with accuracies above 95%.

The Edge Adaptive strategy proves considerably more difficult to detect. At $r = 0.25$, the AUC drops to 0.761 and accuracy falls to 72.5%, with 419 false

positives and 405 false negatives out of the tested pairs. Performance improves substantially at higher rates, reaching an AUC of 0.997 and accuracy of 98.1% at $r = 0.75$.

The Matching LSB strategy is effectively undetectable by RS analysis. AUC values remain near 0.5 across all embedding rates (0.516, 0.504, and 0.514 for $r = 0.25$, 0.50, and 0.75, respectively), indicating performance no better than random guessing. The mean RS estimates for stego images under this strategy (0.038 to 0.044) remain close to the clean image baseline of 0.0211, explaining the classifier's inability to distinguish between the two classes.

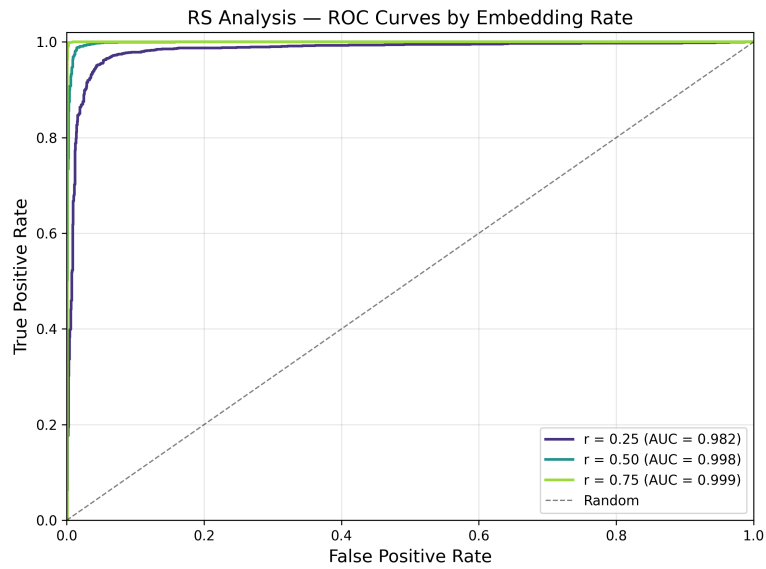


Figure 4.1: RS Analysis: ROC Curves for Sequential LSB embedding at various rates.

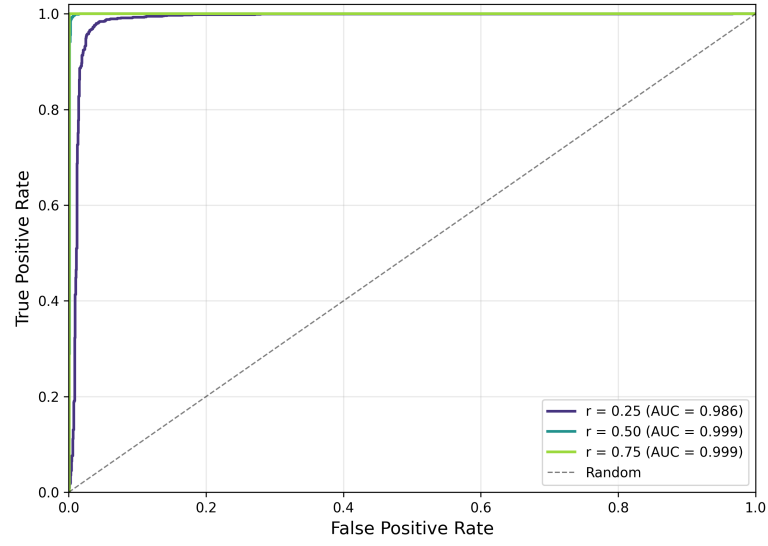


Figure 4.2: RS Analysis: ROC Curves for Randomized LSB embedding at various rates.

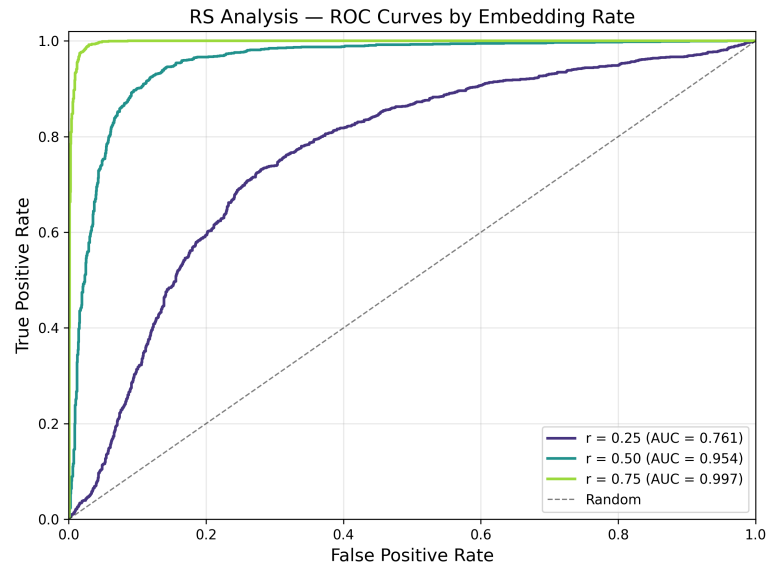


Figure 4.3: RS Analysis: ROC Curves for Edge Adaptive LSB embedding at various rates.

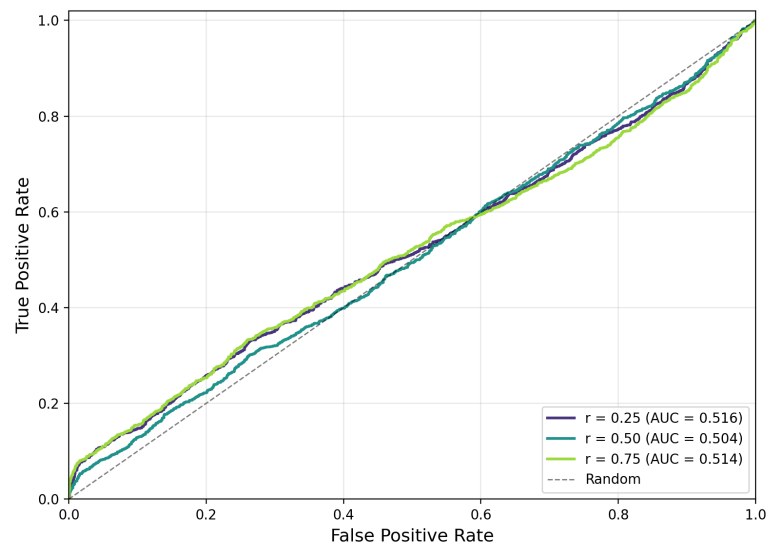


Figure 4.4: RS Analysis: ROC Curves for Matching LSB embedding at various rates.

The corresponding ROC curves are shown in Figures 4.1–4.4. Figures 4.1 and 4.2 illustrate the strong separability achieved for Sequential and Random embedding, with curves closely hugging the upper-left corner. Figure 4.3 shows the progressive improvement for the Edge Adaptive strategy as the embedding rate increases. Figure 4.4 confirms that the Matching LSB strategy produces ROC curves indistinguishable from the diagonal, consistent with random guessing.

Table 4.2: CNN classification performance for different LSB embedding strategies and rates, evaluated on a balanced held-out test set of 3000 images (1500 cover / 1500 stego) per configuration.

Strategy	Rate	AUC	Accuracy	TP	TN	FP	FN
Sequential LSB	0.25	0.960	89.47%	1340	1344	156	160
	0.50	0.994	96.83%	1453	1452	48	47
	0.75	0.999	99.21%	1488	1488	12	12
Random LSB	0.25	0.905	82.35%	1237	1234	266	263
	0.50	0.978	93.18%	1402	1393	107	98
	0.75	0.996	97.64%	1468	1461	39	32
Adaptive LSB	0.25	0.876	79.12%	1195	1179	321	305
	0.50	0.952	88.41%	1335	1317	183	165
	0.75	0.984	94.73%	1427	1415	85	73
Matching LSB	0.25	0.752	68.59%	1042	1016	484	458
	0.50	0.898	82.07%	1245	1217	283	255
	0.75	0.970	91.36%	1380	1361	139	120

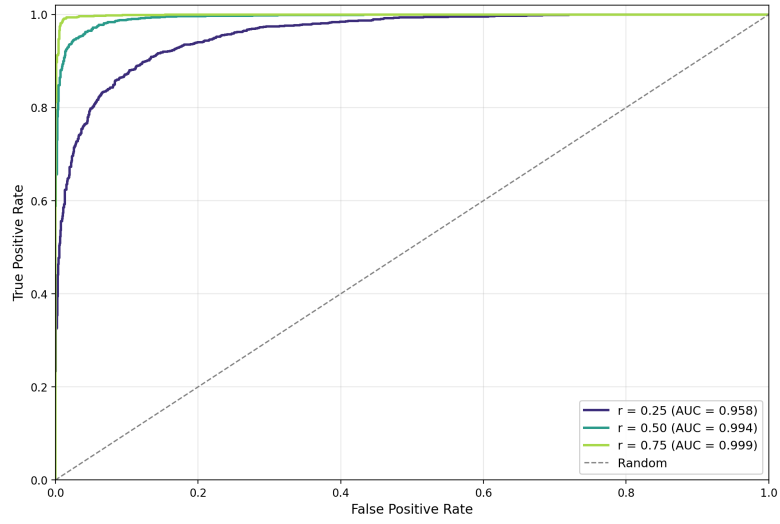


Figure 4.5: CNN: ROC Curves for Sequential LSB embedding at various rates.

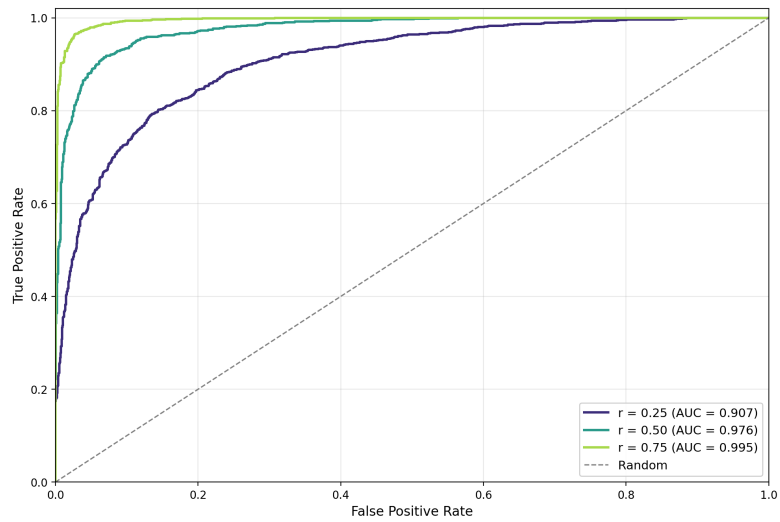


Figure 4.6: CNN: ROC Curves for Randomized LSB embedding at various rates.

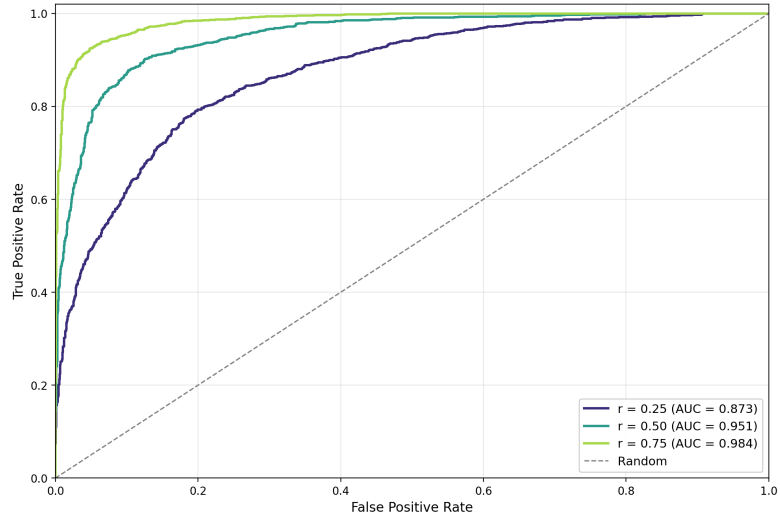


Figure 4.7: CNN: ROC Curves for Edge Adaptive LSB embedding at various rates.

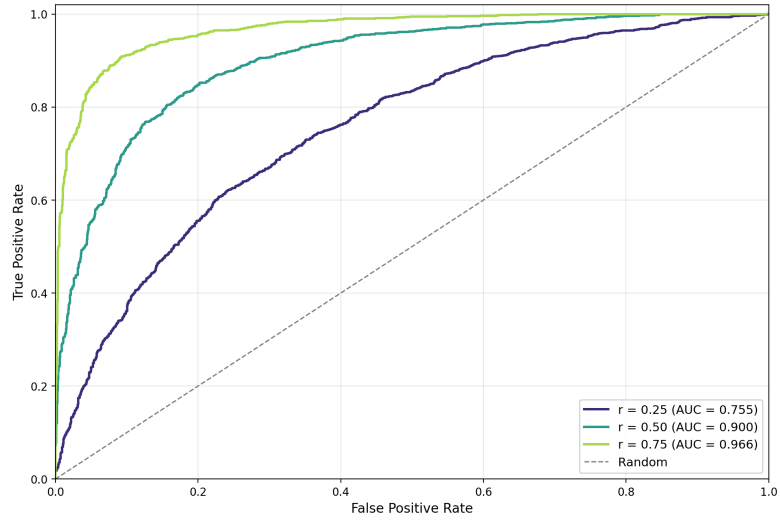


Figure 4.8: CNN: ROC Curves for Matching LSB embedding at various rates.

Accuracy for detection by the trained CNN algorithm is shown in Table 4.2. It is evident from the table that there is a definite hierarchy of detectability: the Sequential LSB method is detected easily, while the Random LSB and Edge Adaptive LSB methods are detected moderately well, with Matching LSB being the most difficult to detect.

In the case of Sequential LSB embedding, the CNN obtains 89.47% accuracy at the smallest embedding rate of $r = 0.25$, with an increase to 96.83% at $r = 0.50$, and 99.21% at $r = 0.75$. This high performance should not come as a surprise, since sequential embedding focuses on changes within contiguous blocks of pixels, resulting in highly structured and spatially correlated errors.

Detecting Random LSB embedding is slightly harder, with an average drop in accuracy of about 7 percentage points compared to the sequential algorithm for $r = 0.25$, where the accuracy falls to 82.35%. For larger values of the distortion parameter, this difference becomes smaller, reaching 93.18% for $r = 0.50$ and 97.64% for $r = 0.75$.

The Edge Adaptive embedding technique makes detection even more difficult. With $r = 0.25$, the accuracy drops to 79.12%, representing a decrease of more than 10 percentage points from Sequential LSB at the same ratio. The accuracy increases to 88.41% and 94.73% for $r = 0.50$ and $r = 0.75$, respectively. With Edge Adaptive embedding, because most alterations are applied to texture-rich areas where natural variations occur, it becomes significantly more difficult for the model to distinguish artificial alterations from the original variance in the image.

Matching LSB proved to be the most challenging method for the CNN to detect. The CNN achieves an accuracy rate of 68.59% when $r = 0.25$. This accuracy rate remains notable, considering it is still significantly better than the chance accuracy rate of 50%. The highest accuracy rates attained when $r = 0.50$ and $r = 0.75$ were 82.07% and 91.36%, respectively. LSB Matching presents the greatest challenge to the CNN because of the basic premise that distinguishes it from other schemes: it adjusts pixel values by randomly adding or subtracting 1 rather than strictly replacing the LSB.

For all four methods, an observable trend exists: accuracy improves alongside the embedding rate, with the most substantial gains observed from $r = 0.25$ to $r = 0.50$ (a gain of 7–14 percentage points), whereas the gain between $r = 0.50$ and $r = 0.75$ is smaller (3–9 percentage points). This diminishing return can be attributed to the non-linear relationship between

the embedding rate and the statistical visibility of the artifacts; doubling the amount of hidden data starting from a very low value provides exponentially more evidence to the model than doubling it from an already saturated state.

5

Discussion

5.1 Interpretation of Results

The main research question asked how a Convolutional Neural Network compares to RS analysis in accurately and reliably detecting LSB steganography in PNG images. Taken as a whole, the results do not support a simple narrative that “deep learning is better”. Both methods demonstrate distinct strengths, and the preferable method heavily depends on the specific embedding strategy and payload size being detected.

For Sequential and Random LSB embedding, the RS analysis performed better at every tested rate. At $r = 0.25$, RS showed 95.4% accuracy on Sequential and 97.0% on Random embedding, while the CNN scored 89.47% on Sequential and 82.37% on Random. The most significant difference between the two steganalysis methods is observed at low-payload embeddings (an advantage of roughly 15 percentage points in favor of RS), which gradually decreases at higher rates. These results perfectly align with the theoretical foundation of RS analysis, which states that Sequential and Random LSB replacement violate the natural balance between the number of pixels in R_M and R_{-M} groups, as established by Fridrich, Goljan, and Du [11]. This demonstrates that RS analysis is a highly optimized detector for the characteristic features of this specific image manipulation, and that generic CNNs offer no meaningful advantage over it in these cases.

The case for Edge Adaptive LSB, however, is very different. Both methods perform well at $r = 0.75$ (RS = 98.1%, CNN = 94.73%), but at $r = 0.25$, RS accuracy falls sharply to 72.5% (AUC 0.761), while the CNN retains a 79.13% accuracy. The adaptive technique’s core mechanism—restricting modifications to textured, noisy regions—is the exact assumption violation that weakens RS analysis. Because the mathematical discrimination function already registers high values in those noisy regions, LSB flips no longer reliably push pixel groups from Regular to Singular, or vice versa. The CNN learns directly from the data distribution rather than relying on an explicit smoothness assumption, making it far less susceptible to this evasion tactic.

The crossover between the two methods on Edge Adaptive LSB embedding thus illustrates the theoretical argument made by Qian et al. [26] and Apau et al. [1]: the rigidity of handcrafted statistical features becomes a critical liability as embedding techniques begin to respect and mimic natural image statistics.

The most stark contrast can be observed in the tests on LSB Matching. Here, the RS analysis operates very closely to flipping a coin; with AUCs of 0.516, 0.504, and 0.514, it is virtually a random classifier. This is not a surprising outcome, as Ker [19] observed that the ± 1 adjustment in LSB Matching does not flip bits deterministically, but rather disturbs pixel values symmetrically. This process does not break the $R_M \approx R_{-M}$ symmetry assumption that RS analysis inherently relies upon. Conversely, the CNN accuracy ranged from 68.60% at $r = 0.25$ to 91.37% at $r = 0.75$. While 68.60% accuracy is not flawless, it is significantly better than a random guess and demonstrates that the neural network can learn to identify artifacts that render RS analysis entirely blind. For the Matching LSB method, this comparison demonstrates which method is applicable at all, rather than merely which method performs better.

5.2 Sub-Question: Sensitivity to Payload Size

Our sub-question asked how detection performance varies as the hidden payload decreases. A degradation in performance can be observed across both methods, but the specific trajectory of that degradation is highly informative.

The degradation of RS analysis is heavily dependent on the embedding technique used to hide the payload. It performs exceptionally well for Sequential and Random embedding, as the drop in accuracy from $r = 0.75$ to $r = 0.25$ is only about four to five percentage points. Thus, RS analysis proves to be highly reliable even at the lowest tested embedding rates for these schemes.

In the case of Adaptive embedding, this same decrease in payload results in a much more severe drop in performance (from 98.1% to 72.5%). For Matching embedding, there is virtually no impact from varying the payload parameter, simply because the detector fails to identify the signal at any rate. This directly correlates with our findings discussed in Section 2.1.4, where the performance of statistical tests decreases when the induced asymmetry falls below the natural noise level of the image.

The CNN is much more consistent across degradation rates, exhibiting an average decrease of 7–14 percentage points from $r = 0.50$ to $r = 0.25$, and a smaller decrease from $r = 0.75$ to $r = 0.50$. Most importantly, there is no catastrophic failure in any embedding scheme. Even on Matching LSB, where RS fails completely at $r = 0.25$, the CNN maintains discriminative capacity. This supports the claim by Apau et al. [1] that learned representations generalize better to low-payload embeddings, even if, in this specific study, that advantage is only explicitly demonstrated against the two more advanced schemes.

Overall, the two approaches provide different answers to our sub-question depending on the embedding scheme. For Sequential and Random LSB, both approaches can be trusted to work down to $r = 0.25$, though RS is slightly more accurate. On Adaptive LSB, the CNN exhibits less catastrophic degradation, making it the preferred detector below $r = 0.50$. Finally, for Matching LSB, only the CNN shows any measurable degradation at all, as RS cannot distinguish the signals at any tested rate.

5.3 Relation to Previous Work

Three common assertions in existing steganalysis research are reaffirmed by these results. First, as established by Fridrich, Goljan, and Du [11] and Fridrich and Goljan [10], RS analysis is a highly effective means of identifying classical LSB replacement, remaining robust across varying rates of embedded data. Second, as predicted by Ker [19], RS-type detectors

fail against LSB Matching due to an inherent lack of structural compatibility with the embedding mechanics. Third, as suggested by Qian et al. [26], Wang, Pan, and Fan [31], and Apau et al. [1], CNN-based steganalysis methods generalize much better across diverse embedding techniques, overcoming the limitations of rigid, handcrafted statistical tests.

This study confirms that the traditional steganalysis approach (RS) is highly successful within its intended operating range, maintaining superiority over neural approaches for basic spatial embeddings. Therefore, practitioners may confidently rely on RS analysis to detect plain LSB replacement in loss-less PNGs, while reserving CNN methods for scenarios involving complex, adaptive, or matching-based embedding schemes.

5.4 Limitations

Numerous limitations apply to the conclusions drawn by this study, specifically regarding the reliance on a single cover source (BOSSBase 1.01) and a single image modality (grayscale images re-encoded as PNGs). Sedighi, Fridrich, and Cogranne [27] found that the statistical characteristics of cover images significantly influence both embedding and detection performance. Therefore, the absolute accuracies obtained in this research should not be directly extended to other cover sources—such as consumer photographs or screen captures—without further empirical validation.

Furthermore, the payload rates examined in this research are relatively high compared to contemporary steganalysis studies (which often evaluate payloads down to 0.1 bpp or lower). It would be highly relevant to observe how performance degradation occurs for these two methods when utilizing significantly smaller payloads (i.e., < 0.05 bpp), and to determine whether the relative advantage of CNNs becomes even more pronounced at these near-invisible payload sizes across varied cover sources.

5.5 Future Work

The limitations mentioned above naturally lead to three primary avenues for future research. The first avenue is to expand the range of payloads down to the sub-0.1 bpp range, allowing researchers to evaluate whether the degradation of CNNs remains graceful at extremely low embedding rates compared to statistical methods. The second avenue is to broaden the set of cover sources to include JPEGs decompressed to PNGs, smartphone

photographs, and synthetically generated images. This would directly address the cover source dependency highlighted by Sedighi, Fridrich, and Cogranne [27] and provide a more stringent assessment of the CNN generalization hypothesis in real-world domains. The third avenue involves expanding the comparison to include highly modern, content-adaptive approaches such as S-UNIWARD and MiPOD. Aligning future comparisons with contemporary steganalysis benchmarking procedures will ultimately help determine whether traditional RS analysis retains any practical utility in modern contexts, or if the field must rely entirely on neural network detectors.

5.6 Broader Impact

The major implication of these findings is that the choice of methodology in applied steganalysis must be dictated by the specific threat model, rather than defaulting to the most “in vogue” deep learning method available. RS analysis remains a powerful, highly interpretable, and computationally efficient detector for plain LSB replacement. Its superior performance against low and high payloads in this analysis firmly endorses its continued use within forensic pipelines targeting basic steganographic content.

However, when the target is an embedding created using an adaptive method or LSB Matching, a machine learning approach must be adopted. The CNN findings in this analysis demonstrate that deep learning can recover usable signals in contexts where RS analysis is completely blind, even if the model’s accuracy falls slightly below the upper bounds reported in literature that utilizes massive computational resources and highly complex architectures. Ultimately, these two methods should not be viewed strictly as competitors, but as complementary tools. A highly viable real-world detection strategy would involve running both methods in parallel and ensembling the results to cover a broader spectrum of steganographic threats.

5.7 Ethical Considerations

These empirical findings allow for a nuanced ethical reflection. The results confirm that the simple, computationally inexpensive RS analysis remains highly sufficient against foundational techniques such as Sequential and Random LSB. This means that forensic practitioners and law enforcement do not necessarily need access to specialized, expensive hardware or

massive training datasets to address a significant share of basic real-world cases.

However, the results also highlight potential exploits by adversaries utilizing Matching LSB, which both RS analysis and our CNN struggled to detect reliably at low payloads ($r = 0.25$). We consider the disclosure of these vulnerabilities acceptable because the embedding strategies examined have been long-established in public literature [28, 19, 22], and this study contributes no novel evasion methods. This research utilizes a standard academic dataset (BOSSBase 1.01) and strictly adheres to the principles of scientific integrity and defensive security research outlined in Chapter 3.

Bibliography

- [1] Richard Apau et al. “Image steganography techniques for resisting statistical steganalysis attacks: A systematic literature review”. In: *PLOS ONE* 19.9 (2024), pp. 1–47. DOI: 10.1371/journal.pone.0308807.
- [2] Patrick Bas, Tomáš Filler, and Tomáš Pevný. “Break our steganographic system: The ins and outs of organizing BOSS”. In: *Information Hiding: 13th International Conference, IH 2011, Prague, Czech Republic, May 18-20, 2011, Revised Selected Papers 13*. Springer. 2011, pp. 59–70.
- [3] Massoud Boroumand, Meng Chen, and Jessica Fridrich. “Deep residual network for steganalysis of digital images”. In: *IEEE Transactions on Information Forensics and Security*. Vol. 14. 5. 2019, pp. 1181–1193. DOI: 10.1109/TIFS.2018.2871749.
- [4] Chi-Kwong Chan and L. M. Cheng. “Hiding data in images by simple LSB substitution”. In: *Pattern Recognition* 37.3 (2004), pp. 469–474. DOI: 10.1016/j.patcog.2003.08.007.
- [5] Samir Chatterjee and Alan Hevner. *Design Research in Information Systems: Theory and Practice*. Berlin: Springer US, 2010.
- [6] Martyn Denscombe. *The Good Research Guide: For Small-Scale Social Research Projects*. 5th. Maidenhead: Open University Press, 2014.
- [7] David R. Eaton. “Steganalysis and the perpetuation of crime using steganography”. Order No. 10251766. Available from ProQuest One Academic; Social Science Premium Collection. (1865313431). PhD thesis. ProQuest Dissertations Publishing, 2016. URL: <https://ezp.sub.su.se/login?url=https://www.proquest.com/dissertations-theses/steganalysis-perpetuation-crime-using/docview/1865313431/se-2>.
- [8] Eltyeb E. Abed Elgabar. “Comparison of LSB Steganography in BMP and JPEG Images”. In: *International Journal of Soft Computing and*

- Engineering (IJSCE)* 3.5 (2013). Retrieval Number: E1900113513/2013, pp. 91–95. ISSN: 2231-2307.
- [9] Numrena Farooq and Roohie Naaz Mir. *Image Steganalysis Using Deep Convolution Neural Networks: A Literature Survey*. Vol. 14. 4. Bentham Science Publishers, 2024, pp. 247–264. DOI: 10.2174/0122103279296370240529075507.
 - [10] Jessica Fridrich and Miroslav Goljan. “Practical steganalysis of digital images: State of the art”. In: *Proceedings of SPIE - Security and Watermarking of Multimedia Contents IV*. Ed. by Edward J. Delp and Ping W. Wong. Vol. 4675. SPIE, 2002, pp. 1–13. DOI: 10.1117/12.465263.
 - [11] Jessica Fridrich, Miroslav Goljan, and Rui Du. “Reliable Detection of LSB Steganography in Color and Grayscale Images”. In: *Proceedings of the 2001 Workshop on Multimedia and Security: New Challenges*. ACM, 2001, pp. 27–30. DOI: 10.1145/1232454.1232466.
 - [12] Jessica Fridrich et al. “Quantitative steganalysis of digital images: Estimating the secret message length”. In: *Multimedia Systems* 9.3 (2003), pp. 288–302. DOI: 10.1007/s00530-003-0100-9.
 - [13] Naveen Goud. *The growing cyber threat of steganography: Concealing malicious activity in plain sight*. Accessed: 2025-11-15. 2025. URL: <https://www.cybersecurity-insiders.com/the-growing-cyber-threat-of-steganography-concealing-malicious-activity-in-plain-sight/>.
 - [14] Martin Gullbrandsson and Jan Pakos. *RS-Analysis Implementation*. 2026. URL: <https://github.com/Gullbra/RS-analysis>.
 - [15] Martin Gullbrandsson and Jan Pakos. *SRNet Implementation*. 2026. URL: <https://github.com/Gullbra/Pytorch-implementation-of-SRNet>.
 - [16] Martin Gullbrandsson and Jan Pakos. *Steganalysis Project*. 2026. URL: <https://github.com/Gullbra/ThesisWorkBachelor>.
 - [17] Nurul Hamid et al. “Image steganography techniques: An overview”. In: *International Journal of Computer Science and Security* 6.3 (2012), pp. 168–187. URL: <https://www.cscjournals.org/library/manuscriptinfo.php?mc=IJCSS-670>.
 - [18] Vojtěch Holub, Jessica Fridrich, and Tomáš Denemark. “Universal distortion function for steganography in an arbitrary domain”. In: *EURASIP Journal on Information Security* 2014.1 (2014), pp. 1–13.
 - [19] Andrew D. Ker. “Steganalysis of LSB matching in grayscale images”. In: *IEEE Signal Processing Letters* 12.6 (2005), pp. 441–444. DOI: 10.1109/LSP.2005.847889.

- [20] Diederik P. Kingma and Jimmy Ba. “Adam: A Method for Stochastic Optimization”. In: *arXiv* (2014). DOI: 10.48550/arxiv.1412.6980.
- [21] Baoxun Li et al. “A survey on image steganography and steganalysis”. In: *Journal of Information Hiding and Multimedia Signal Processing* 2.2 (2011), pp. 142–172.
- [22] Weiqi Luo, Fangjun Huang, and Jiwu Huang. “Edge adaptive image steganography based on LSB matching revisited”. In: *IEEE Transactions on Information Forensics and Security* 5.2 (2010), pp. 201–214. DOI: 10.1109/TIFS.2010.2041812.
- [23] Sathiamoorthy Manoharan. “An Empirical Analysis of RS Steganalysis”. In: *2008 The Third International Conference on Internet Monitoring and Protection*. IEEE, 2008, pp. 1–5. DOI: 10.1109/ICIMP.2008.15.
- [24] Anupam Nag et al. “Semi random position based steganography for resisting statistical steganalysis”. In: *International Journal of Network Security* 17.1 (2015), pp. 57–65.
- [25] Tomáš Pevný, Tomáš Filler, and Patrick Bas. “Using high-dimensional image models to perform highly undetectable steganography”. In: *Information Hiding: 12th International Conference, IH 2010, Calgary, AB, Canada, June 28-30, 2010, Revised Selected Papers 12*. Springer, 2010, pp. 161–177.
- [26] Yinlong Qian et al. “Deep Learning for Steganalysis via Convolutional Neural Networks”. In: *Media Watermarking, Security, and Forensics 2015*. Vol. 9409. SPIE, 2015, 94090J. DOI: 10.1117/12.2083479.
- [27] Vahid Sedighi, Jessica Fridrich, and Rémi Cigrange. “Toss that BOSS-base, Alice!” In: *IS&T International Symposium on Electronic Imaging*. San Francisco, United States: Society for Imaging Science and Technology, 2016.
- [28] Tristan Sharp. “An Implementation of Key-Based Digital Signal Steganography”. In: *Information Hiding. IH 2001*. Ed. by Ira S. Moskowitz. Vol. 2137. Lecture Notes in Computer Science. Springer, 2001, pp. 223–234. DOI: 10.1007/3-540-45496-9_2.
- [29] G. L. Smitha and E. Baburaj. “Sobel edge detection technique implementation for image steganography analysis”. In: *Biomedical Research* 29.Special Issue (2018). DOI: 10.4066/biomedicalresearch.29-17-1212.
- [30] *Usage statistics of image file formats for websites*. Last updated 5 December 2025. Q-Success. 2025. URL: https://w3techs.com/technologies/overview/image_format (visited on 12/06/2025).

- [31] Haihua Wang, Xiaohui Pan, and Ling et al. Fan. “Steganalysis of convolutional neural network based on neural architecture search”. In: *Multimedia Systems* 27 (2021), pp. 379–387. DOI: 10 . 1007 / s00530 - 021 - 00779 - 5.
- [32] Andreas Westfeld and Andreas Pfitzmann. “Attacks on steganographic systems”. In: *Information Hiding: IH 1999*. Ed. by Andreas Pfitzmann. Vol. 1768. Lecture Notes in Computer Science. Springer, 2000, pp. 61–76. DOI: 10 . 1007 / 10719724 _ 5.